

The increase in cell volume and nuclear number of the koji-fungus *Aspergillus oryzae* contributes to its high enzyme productivity

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The ratio of nuclei to cell volume is a well-controlled parameter in eukaryotic cells. This study now reports **important** findings that expand our understanding of the regulatory relationship between cell size and number of nuclei. The evidence supporting the conclusions is **convincing** obtained by applying appropriate and validated methodology in line with current state-of-the-art. The paper will be of broad interest for cell biologists and fungal biotechnologists seeking to understand mechanisms determining cell size and number of nuclei and why this knowledge might also be of importance for the production of enzymes and thus production strains not only of *Aspergillus oryzae* but also other industrially used fungi.

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Abstract

While the ratio of nuclei to cell volume is well regulated, it remains largely unexplored in multinucleate organisms. The koji-fungus *Aspergillus oryzae*, traditionally used in Japanese brewing and fermentation for over a thousand years, is now widely utilized in modern biotechnology as a host for enzyme production. We discovered that, over time in culture, hyphae become thicker, resulting in a tenfold increase in cell volume, and the number of nuclei in hyphal cells also increases tenfold, exceeding 200. The increase in cell volume and nuclear number is unique among the investigated *Aspergillus* species and correlates with its high enzyme production capabilities. Since nuclear number and cell volume are correlated, both must increase simultaneously. Our analyses identified genetic factors involved in each

of these increases. Increases in nuclear number and cell volume were also observed in other fungi bred for industrial use. This study not only deepens our understanding of the evolutionary processes that promote high enzyme productivity through fungal breeding, but also provides insights into the molecular mechanisms regulating cell volume and nuclear number in multinucleate organisms.

Introduction

In unicellular organisms, cell volume is tightly regulated in coordination with nuclear division timing (1–3). Similarly, in multinucleated filamentous fungi, a correlation exists between cell volume and nuclear number (4, 5). However, the number of nuclei varies across species (6), the factors governing nuclear number and cell volume remain largely unknown. Filamentous fungi are industrially significant microorganisms, and the strains selected through breeding provide ideal research models for studying these evolutionary processes.

Filamentous fungi secrete a variety of hydrolytic enzymes to support their growth, and humans have long harnessed this enzyme secretion ability for the production of a wide range of fermented foods and beverages (7). *Aspergillus oryzae* has been used in Japan for over a thousand years to produce traditional fermented foods like sake, soy sauce, and miso (8, 9). A distinctive feature of the fermentation techniques is solid-state cultivation using substrates like rice, soybeans, and wheat bran. The inoculum of *A. oryzae* used in fermentation, known as “koji,” has been commercially produced for around 700 years (10). Koji is created by cultivating koji-fungus on grains, and koji starter manufacturers have selected and bred strains to improve the flavor and color of various fermented products. The *A. oryzae* strains currently in use are mainly preserved and managed by koji manufacturers and research institutions. A notable characteristic of *A. oryzae* is its high ability to secrete starch-degrading enzymes and proteases, which has led to extensive research into its genome and gene expression regulation (11–17).

Although *A. oryzae* shares 99.5% genomic similarity with *Aspergillus flavus*, which produces the carcinogenic aflatoxin (18), there have been no reports of *A. oryzae* producing aflatoxins, and its safety has been confirmed at the molecular level (19, 20). Phylogenomic analyses suggest that *A. oryzae* and *A. flavus* diverged approximately 50,000 to 189,000 years ago (21, 22). Over the following millennia, human use of *A. oryzae* is thought to have driven significant genomic recombination, resulting in its evolution into a “cell factory” specifically optimized for the breakdown of sugars and proteins (23).

Additionally, modern biotechnology utilizes filamentous fungi as cell factories to produce organic acids, enzymes, and pharmaceuticals (24–26). *A. oryzae* also shows high production capacity as a host for both homologous and heterologous protein production in modern biotechnology (27, 28). Recently, its application has expanded to include the production of pharmaceutical proteins and secondary metabolites (29, 30). Furthermore, filamentous fungi involved in fermentation play a crucial role in creating a more sustainable food system, such as by upcycling agricultural by-products into food through fungal fermentation and using *A. oryzae* mycelium for alternative meats (31, 32).

While basic and applied research on *A. oryzae* continues, fundamental questions remain unanswered, such as why *A. oryzae* exhibits high enzyme production capacity and why it excels as a host for heterologous expression. This study shows that the key lies in the increase in cell volume and nuclear number, and analyzes the molecular mechanisms.

Result

Increase in nuclear number and cell volume in *Aspergillus oryzae*

In the model strain *A. oryzae* RIB40, hyphae on day 1 of cultivation contain 10–20 nuclei per cell, but by day 2 of cultivation, apical cells with more than 200 nuclei appear (33). We classified the hyphae into three categories based on the number and distribution of nuclei, and compared their proportions after 1 to 3 days of cultivation (see methods, images were shown previously)(33). We found that the proportion of hyphae with a higher number of nuclei (represented by class III, over 200 nuclei in the cell) increased as the culture period increased from 1 to 3 days (Fig. 1A, B) (33). This phenotype was not observed in the model fungus *Aspergillus nidulans* or the closely related species *A. flavus* (Fig. 1A, B) (33). As far as we know, such phenotypes have not been observed in other strains of *Aspergillus nidulans*, *Aspergillus niger*, *Aspergillus fumigatus*, or other strains commonly used in research. Among other *Aspergillus* species used in fermentation (9, 10), the proportion of hyphae with a higher number of nuclei increased in *Aspergillus sojae*, which is used for miso and soy sauce production. This phenotype was not observed in *Aspergillus luchuensis* nor in its albino mutant *Aspergillus luchuensis mut. kawachii*, which are used for distilled spirits such as awamori and shochu. Among several strains of *A. oryzae* used in industrial applications, the RIB915 strain used for soy sauce fermentation does not show an increase in the number of nuclei, while the RIB128 and RIB430 strains used in sake brewing exhibit a significant increase in the number of nuclei, suggesting that different phenotypes were selected through distinct breeding processes.

The distribution of nuclei in the hyphae of *A. oryzae* RIB40 was visualized through 3D reconstruction (Fig. 1C, Movie 1). In the class I hyphae, the nuclei were spaced relatively evenly, whereas in the class III hyphae with increased nuclei, the nuclei were densely packed and arranged in a disordered manner. Furthermore, the nuclear size in the class III hyphae was significantly smaller than that in the class I hyphae (Fig. 1D, E). Nuclear division is synchronized in the hypha even when there are more than 200 nuclei (33), suggesting that DNA replication occurs similarly in most nuclei. The germination rate of conidia and the colonies formed from individual conidia show no significant abnormalities, suggesting, that nearly all nuclei possess normal genomes and chromosomes.

Strains without an increase in nuclear number had hyphae with diameters ranging from 2 to 7 μm , while strains with an increase in nuclear number had some hyphae of similar thickness but often exhibited thicker hyphae exceeding 10 μm (Fig. 1F). We compared the number of nuclei and cell volume within the first 100 μm from the hyphal tip, due to variation in the distance from the hyphal tip to the septum, among *A. oryzae* strains (RIB40, RIB128, RIB915) and *A. flavus* (Fig. 1G). A strong positive correlation was observed between number of nuclei and cell volume. This is consistent with previously reports that the number of nuclei per hyphal volume remains constant (4, 5).

Thick hyphae with increased nuclei emerge by branching

Time-lapse imaging using the *A. oryzae* RIB40 expressing H2B-GFP indicated that thick hyphae with increased nuclei emerged by branching from hyphae without increased nuclei (Fig. 2A, Movie 2). In the emerged thick hyphae, rapid nuclear division occurred at the sites of branching (Fig. 2B, Movie 3). Immediately after branching, the thick hyphae showed a significantly faster rate of nuclear proliferation, increasing 4–5 times within 8 hours compared to the thin hyphae immediately after branching (Fig. 2C). In some cases, thick hyphae emerged by branching from thick hyphae (Fig. 2D, left), while in other cases, thin hyphae emerged from thick hyphae (Fig. 2D, right).

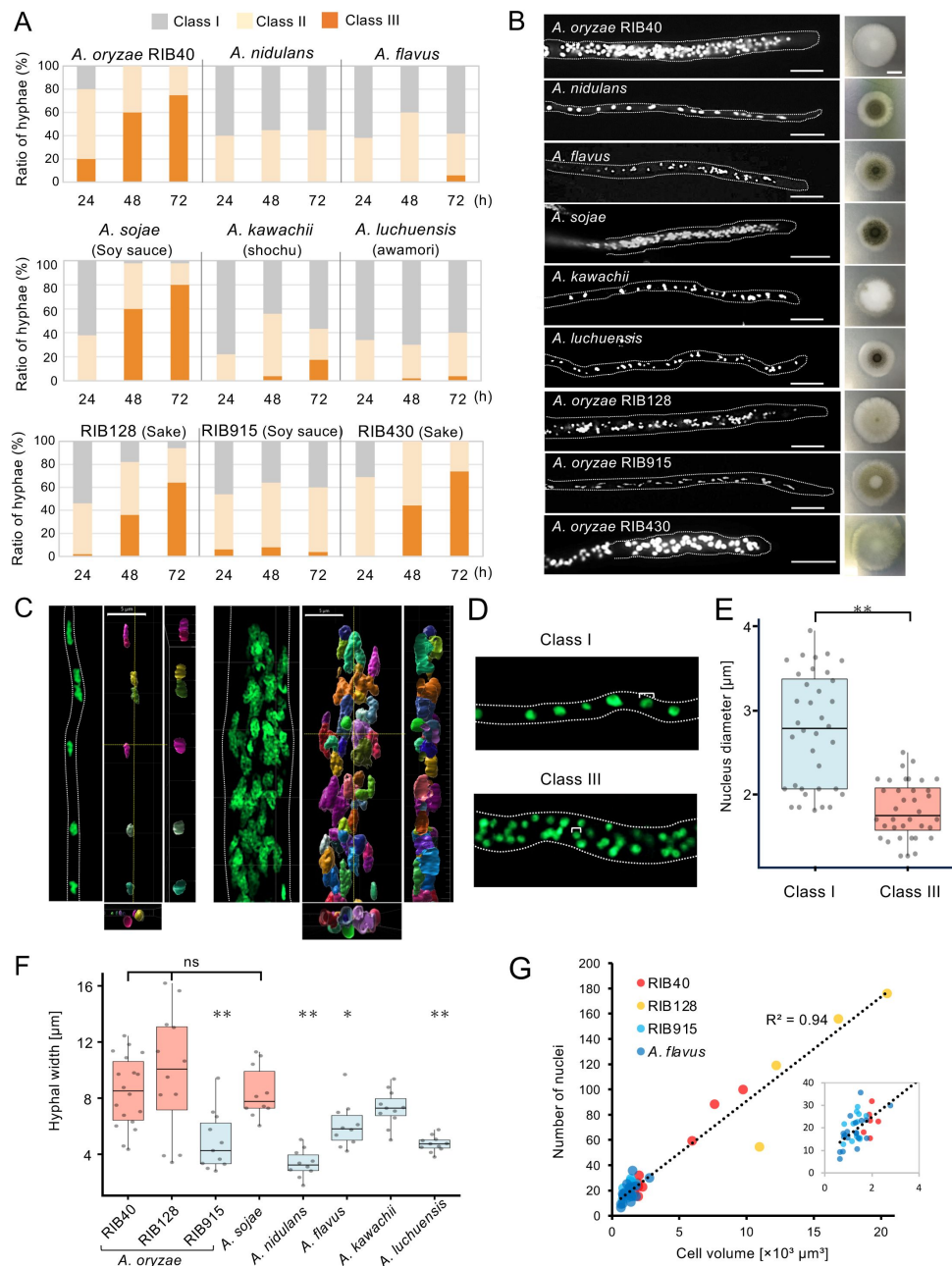


Figure 1.

Increase in number of nuclei and cell volume in *Aspergillus* species.

(A) The nuclear distribution in the tip cells was categorized into Classes I–III. The ratio of hyphae in each class was measured in *Aspergillus* species at 24, 48, and 72 hours of growth ($n=50$). Data for *A. oryzae* RIB40 and *A. nidulans* are reproduced from a previous study (33). (B) Colony morphology after 3 days of culture on the minimal medium. The nuclear distribution in the hyphae at the colony periphery stained with SYBR Green. Scale bars: hyphae, 20 μm ; colonies, 1 cm. (C) 3D images of hyphae without increased nuclei (left) and with increased nuclei (right) in *A. oryzae* RIB40 from Movie 1. Each nucleus is indicated with different colors by Imaris soft. Scale bar: 5 μm . (D) Comparison of nuclear size in Class I hyphae (2.4 μm) and Class III hyphae (1.6 μm), indicated by white lines. (E) Box plots of nucleus diameters in hyphae without increased nuclei and those with increased nuclei ($n=35$, ** $p < 0.01$, t-test). (F) Box plots of hyphal width at the colony periphery grown for 72 hours ($n=10$ –14, ** $p < 0.01$, * $p < 0.05$, t-test). Strains with increased nuclei are marked in red, and those without are marked in blue. (G) Correlation between nucleus number and cell volume of hyphae at 100 μm from the tips in *A. oryzae* RIB40, RIB128, RIB915, and *A. flavus*.

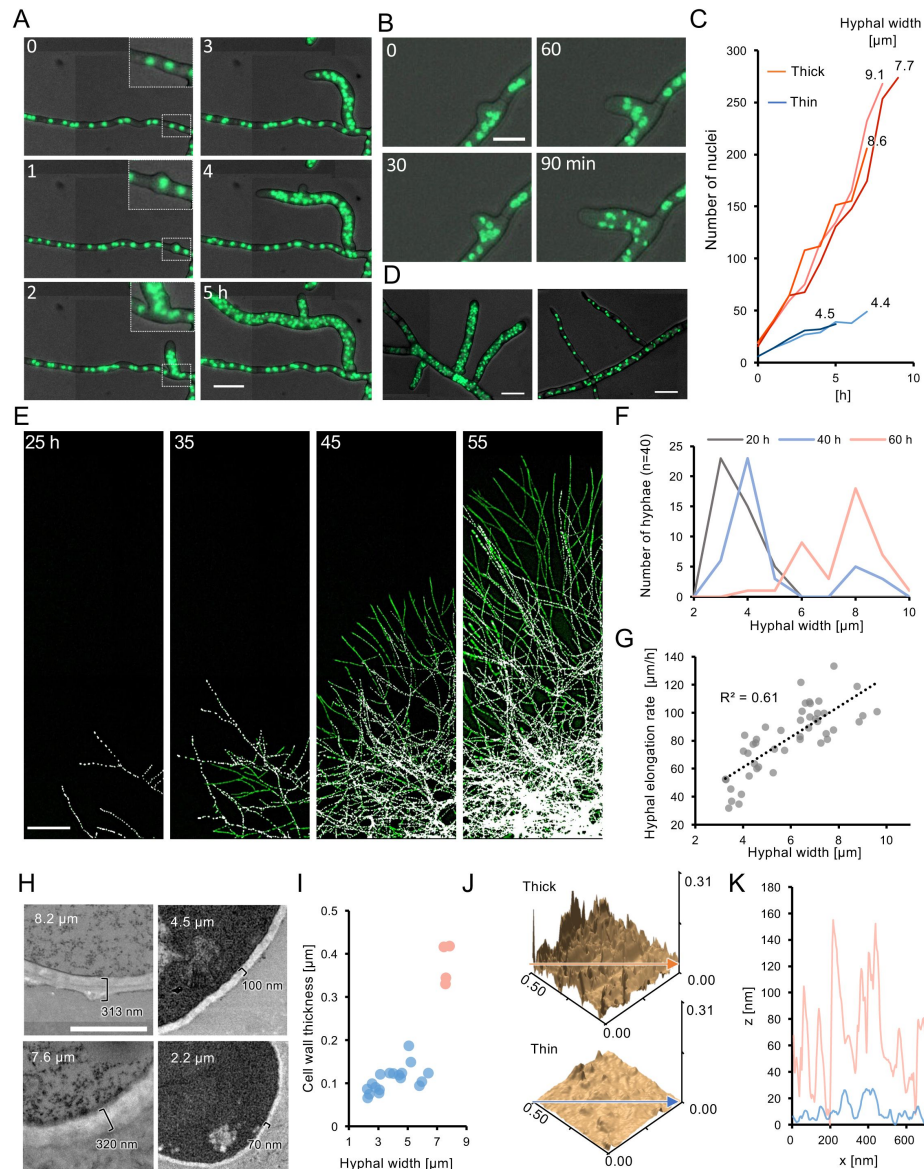


Figure 2.

Thick hyphae with increased nuclei emerge by branching.

(A) Time-lapse image sequence of *A. oryzae* RIB40 expressing H2B-GFP showing the emergence of thick hyphae with increased nuclei by branching from [Movie 2](#). Scale bar: 20 μm. Elapsed time is indicated in min. (B) Image sequence of successive nuclear division within the newly emerged branched hypha from [Movie 3](#). Scale bar: 10 μm. Elapsed time is indicated in min. (C) Time course of nuclear number per hypha. Nuclear numbers were counted with the start time of branching set as 0, every hour for 5 to 10 h. Thick; hyphal width >7 μm, Thin; hyphal width <5 μm. (D) Branching from hyphae with increased nuclei generates both thick hyphae with increased nuclei (left) and thin hyphae without increased nuclei (right). Scale bars: 20 μm. (E) Image sequence of mycelial growth showing hyphae with increased nuclei (green) and without increased nuclei (white) from [Movie 4](#). Scale bars: 200 μm. Elapsed time is indicated in h. (F) Line histogram of hyphal width at the colony periphery at 20, 40, and 60 h, calculated from [Movie 4](#) (n=40). (G) Correlation between hyphal width and maximum elongation rate calculated from [Movie 4](#) (n=45). The maximum elongation rate is determined from elongation rates measured every hour over 10 hours period. (H) TEM images of *A. oryzae* RIB40 hyphae. Hyphal diameter and cell wall thickness are indicated white and black text respectively. Scale bar: 1 μm. (I) Correlation plots of hyphal width and cell wall thickness based on TEM images. Thick hyphae (>7 μm) are marked in red, and thin hyphae (<7 μm) are marked in blue. (J) 3D surface images of *A. oryzae* RIB40 hyphal tips in the thick hypha (upper) and the thin hypha (lower) constructed using AFM. (K) Surface roughness of cell walls in thick (red) and thin (blue) hyphae along the arrows in J.

Time-lapse imaging was conducted over a wider area, and image processing was applied to color-code the nuclei: green for those in hyphae with increased nuclei and white for those in hyphae without increased nuclei (**Fig. 2E**, **Movie 4**, Fig.S1A, see methods). Up to 24 hours after inoculation, most hyphae were thin with no increase in nuclei number, but thick hyphae with increased nuclei began to appear as time progressed (**Fig. 2E, F**). Since thick hyphae elongated faster than thin hyphae (**Fig. 2G**), thick hyphae outgrew and overtook thin hyphae, eventually dominating the colony perimeter (**Fig. 2E, F**).

Transmission electron microscopy confirmed basic nuclear structures, such as the nuclear membrane and nucleolus, in both thick and thin hyphae (Fig. S1B). As hyphal diameter increased, cell walls became more uneven and thicker (**Fig. 2H, I**). This finding corresponds with Raman spectroscopy results showing pronounced peaks corresponding to cell wall polysaccharides in thick hyphae (Fig. S1C)(34). Additionally, peaks associated with active mitochondria were prominent in thick hyphae (Fig. S1C)(35), which was further confirmed by fluorescence staining (Fig. S1D). The surface structure and mechanical properties of the cell wall were analyzed using atomic force microscopy. Thick hyphae exhibited greater surface roughness and higher elasticity compared to thin hyphae (**Fig. 2J, K**, Fig. S1E, F).

Increase in nuclear number and enzyme secretion

We compared the amounts of secreted proteins in *A. oryzae* RIB40, RIB128, RIB915, *A. nidulans*, and *A. flavus* (**Fig. 3A**). The strains that show an increase in nuclear number had significantly higher levels of protein secretion than the strains that do not. In this condition, α -amylases are the dominant proteins secreted by *A. oryzae* (15). The amylase enzyme activity was compared between in *A. oryzae* RIB40, which shows an increase in nuclear number, and *A. oryzae* RIB915, which does not show an increase. In RIB40, starting from day 2, when hyphae with increased nuclear numbers appeared, the enzyme activity per fungal biomass significantly increased more than RIB915 (**Fig. 3B**).

To compare the amylase activity between hyphae with a higher and a lower number of nuclei, amylase activity was monitored under conditions where a single hypha grew in a microfluidic channel with a fluorescent substrate (**Fig. 3C**, S2A, **Movie 5**)(36). After 12 hours, the fluorescence intensity in the channel with thick hyphae was about three times higher than that in the channel with thin hyphae, indicating that thicker hyphae secreted more amylase than thinner hyphae (**Fig. 3D**).

We found that addition of yeast extract increased in both hyphal width and nuclear number in the RIB915 strain (**Fig. 3E**, Fig. S2B). In *A. oryzae* RIB40, the addition of yeast extract did not make the already thick hyphae any thicker but increased the proportion of thicker hyphae. Addition of yeast extract increased protein secretion approximately 1.8 times in RIB40 and 8.5 times in RIB915 (**Fig. 3F**). Although the addition of yeast extract did not cause a dramatic increase in nuclear number in *A. nidulans*, hyphal width increased by 1.4-times and protein secretion increased by 5.1-times.

The addition of nucleic acids, peptone, or casamino acids to the minimal medium increased the number of nuclei in *A. oryzae* RIB915, while vitamin B did not cause a significant change in nuclear number (Fig. S2C). Although the addition of individual amino acids did not show the same effect as yeast extract, the addition of asparagine, proline, glutamine, and branched-chain amino acids increased the proportion of hyphae with increased nuclei by 40-50% (**Fig. 3G**, Fig. S2D). The amino acids that induced nuclear number increase also led to an increase in protein secretion, indicating a positive correlation between the ratio of class III hyphae and the increase in secreted proteins (**Fig. 3H**, Fig. S2E). Since the amino acids inducing an increase in nuclear number may activate the TOR pathway (37), *A. oryzae* RIB40 was grown with low concentration of rapamycin, a TOR pathway inhibitor, which minimally affects colony size. Under the conditions,

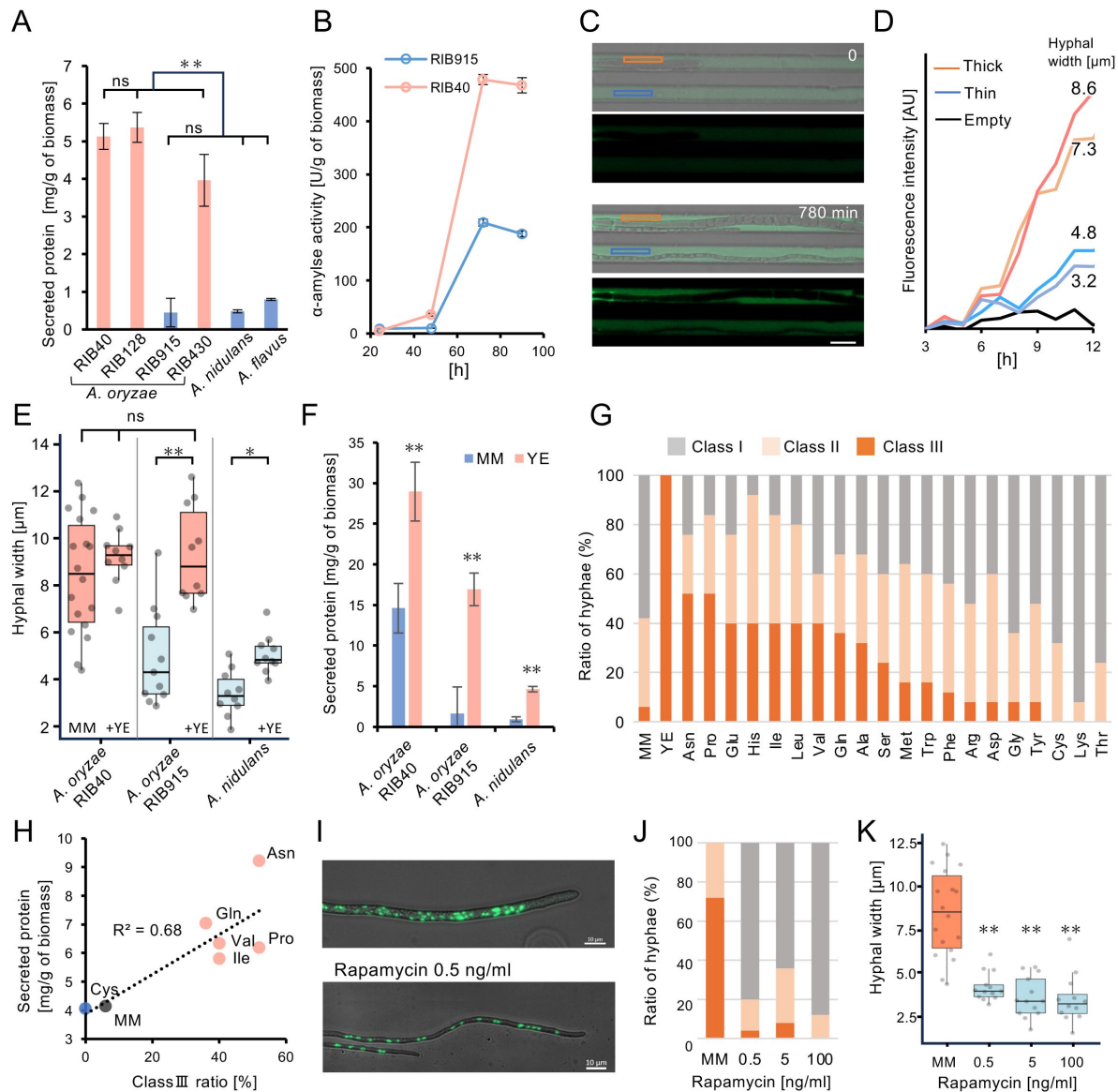


Figure 3.

Correlation between number of nuclei and enzyme secretion.

(A) Secreted protein per biomass in strains with increased nuclei (red) and without increased nuclei (blue) after 4 days of culture in minimal medium with 1% maltose (mean $\pm$ S.E., $n=3$, ** $p < 0.01$, t-test). (B) Time course of α -amylase activity per biomass in *A. oryzae* RIB40 and RIB915 cultured in minimal medium with 1% maltose (mean $\pm$ S.E., $n=3$). (C) Measurement of α -amylase activity in a single hypha using a microfluidic device. A fluorescent substrate increases in fluorescence upon hydrolysis by α -amylase. ROIs for thick hyphae are marked in red, and thin hyphae in blue. Scale bar: 10 μ m. (D) Temporal changes in fluorescence intensity measured in individual flow channels, as described in C. (E) Box plots of hyphal width in colonies grown in minimal medium with or without 1% yeast extract ($n=10-18$, ** $p < 0.01$, * $p < 0.05$, t-test). Conditions with increased nuclei are marked in red, and those without in blue. (F) Secreted protein per biomass in minimal medium with or without 1% yeast extract (mean $\pm$ S.E., $n=3$, ** $p < 0.01$, t-test). (G) Ratio of class I-III hyphae in *A. oryzae* RIB915 colonies grown in minimal medium supplemented with 1% yeast extract or 0.1% individual amino acids ($n=50$). (H) Correlation between the ratio of Class III hyphae and secreted protein per biomass in the 0.1% amino acid-supplemented medium. (I) Images of hyphae and nuclear distribution in *A. oryzae* RIB40 grown on minimal medium with or without 0.5 ng/ml rapamycin. Scale bars: 10 μ m. (J) Ratio of class I-III hyphae in *A. oryzae* RIB40 cultured on minimal medium containing 0, 0.5, 5, or 100 ng/ml rapamycin ($n=50$). (K) Box plots of hyphal width in the colonies cultured under the conditions in J ($n=11-18$, ** $p < 0.01$, t-test).

RIB40 did not produce hyphae with an increased number of nuclei (**Fig. 3I-K**, Fig. S2F). Rapamycin decreased the ratio of hyphae with increased nuclei even in the medium with yeast extract (Fig. S2G).

Transcriptome analyses in hyphae with increased nuclei

To investigate gene expression changes in hyphae with increased nuclei, transcriptome analysis was performed under six conditions: In minimal medium; *A. oryzae* RIB40, which shows increased nuclei, *A. oryzae* RIB915 and *A. nidulans*, which do not show increased nuclei. In minimal medium with yeast extract, *A. oryzae* RIB40 and RIB915, which shows increased nuclei, and *A. nidulans*, which do not show increased nuclei (**Fig. 4A**, Fig. S2B). In RIB915, supplementation with yeast extract led to a more than four-fold increase in the expression of 660 genes, among which 449 genes did not show such increase in RIB40 or *A. nidulans* (**Fig. 4B**, **Table S1**). GO analysis of these genes revealed that process related to cell wall synthesis and divalent metal ion transport were significantly enriched (**Fig. 4C**).

To capture gene expression changes associated with increased nuclei with higher resolution, thick and thin hyphae from RIB40 were dissected using laser microdissection, collected separately, and subjected to transcriptome analysis (**Fig. 4D**, **Movie 6**). Clear differences in gene expression pattern were observed between thick and thin hyphae (**Fig. 4E**). 558 genes that were more than four-fold upregulated in thick hyphae but not in thin hyphae (**Table S2**). GO analysis of the upregulated genes indicated an enrichment of processes related to biogenesis of ribosome and cellular component (**Fig. 4F**). Among them, 21 matched those from **Table S1** (**Fig. 4G**, **Table S3**). Of these, 13 had GO annotations, which were related to conidiation, cell wall synthesis, Ca^{2+} transport and rRNA processing (**Fig. 3H**). KEGG map visualization indicates that overall ribosomal gene expression increased in the thick hyphae than the thin hyphae (Fig. S3A).

Among the gene list, we focused on *msy1* and *msy2*, which are involved in Ca^{2+} transport and maintaining cell volume homeostasis in *Schizosaccharomyces pombe* (38, 39). Disruption of the orthologue genes, *msyA* and *msyB*, in *A. oryzae* RIB40 did not have a significant impact on colony growth, hyphal morphology and number of nuclei in minimal medium (Fig. S3B). Under hypoosmotic shock conditions, the ΔmsyA often indicated cell lysis near the hyphal tips (**Fig. 4I, J**). Under hypoosmotic conditions, water influx into the cells causes hyphal cell expansion, but the volume and turgor pressure are regulated to prevent cell lysis. The *msyA* disruption, however, impairs the regulation of the hyphal cell expansion, leading to hyphal lysis (Fig. S3C-E).

SNP analysis among industrial strains with distinct phenotypes

The fact that currently used *A. oryzae* strains are managed and preserved by koji manufacturers and research institutions is a notable advantage of *A. oryzae* research. Whole-genome sequencing and comparative genome analysis of 82 *A. oryzae* strains with different applications have shown that these strains cluster into several clades (24). We investigated whether nuclear numbers increase in 20 strains selected from six different clades (**Fig. 5A**). In the minimal media supplemented with yeast extract, all strains exhibited an increase in nuclear number (Fig. S4A). In the minimal media without yeast extract, three strains from Clade C showed an increase in nuclear number, whereas strains from Clades A, B, and E did not (**Fig. 5A, B**, Fig. S4A). Clades F and G included both strains with increased nuclear numbers and strains without. SNP analysis of ORFs in Clades F and G revealed 108 mutations in Clade F between TK-32 and TK-38, and 23 mutations in Clade G between TK-41 and TK-47 (**Table S4** and **S5**).

Among these, AO090038000626; *rseA*, predicted glycosyl transferase, was the only common gene. This gene has been identified as a mutation in a *A. sojae* mutant with high extracellular enzyme production (40). It is also shown that deletion of *rseA* in *A. nidulans* leads to cell wall defects and promotes enzyme secretion (40). Comparison of mutations in *rseA* among strains in Clade C, F

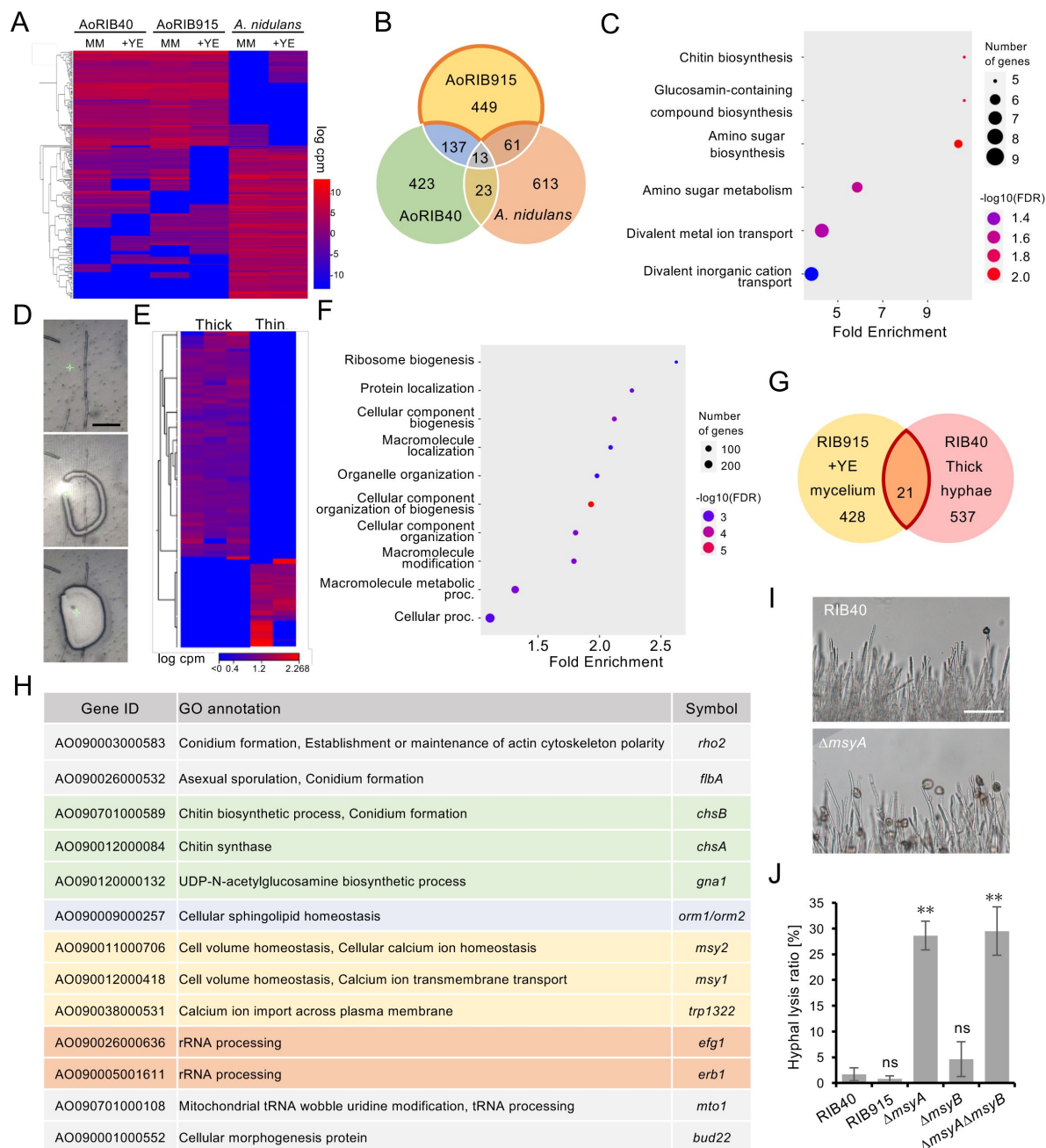


Figure 4.

Transcriptome analyses in hyphae with increased nuclei.

(A) Heatmap of gene expression in *A. oryzae* RIB40, RIB915, and *A. nidulans* grown in the minimal medium with or without 1% yeast extract. The top 500 genes with the largest variation in expression are shown. (B) Venn diagram of genes upregulated more than four-fold in the medium with yeast extract. (C) GO process analysis of genes uniquely upregulated in *A. oryzae* RIB915 grown in the medium with yeast extract. (D) Image sequence of cutting and collecting the targeted hypha by using laser microdissection. Scale bar: 100 μ m. (E) Heatmap of gene expression in thick (n=3) and thin (n=2) hyphae dissected from *A. oryzae* RIB40. The top 300 genes with the largest expression differences are shown. (F) GO process analysis of genes upregulated more than four-fold in thick hyphae compared to thin hyphae. (G) Venn diagram of 449 genes upregulated in RIB915 with yeast extract and 558 genes upregulated in thick hyphae of RIB40. (H) GO annotations of 13 genes from the 21 shared genes in G. Genes related to cell wall processes are shown in green, cell membrane in blue, Ca^{2+} transport in yellow, and rRNA processing in orange. (I) Images of hyphae of RIB40 and Δ *msyA* strains after 3 min of low osmotic stress. Scale bar: 200 μ m. (J) Ratio of hyphal tip lysis under the hypoosmotic shock (mean $\pm$ S.E., n=50, ** p < 0.01, t-test).

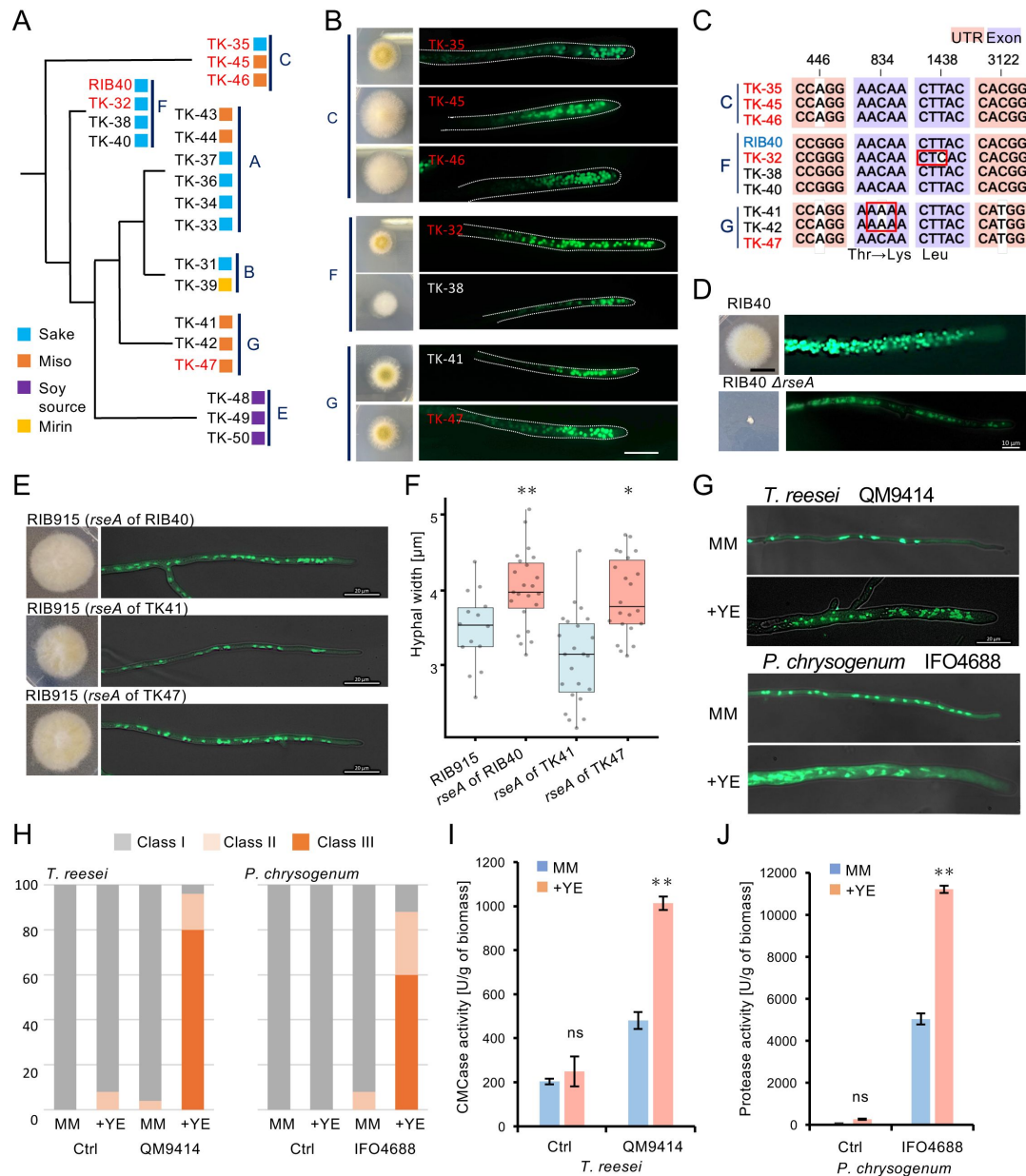


Figure 5.

Comparison of industrial bred strains in *A. oryzae*, *T. reesei* and *P. chrysogenum*.

(A) Phylogenetic clade A-F of *A. oryzae* TK strains. The strains with increased nuclei are indicated in red. (B) Colonies and hyphae stained with SYBR Green of strains in clades C, F, and G grown for 3 days on the minimal medium. Scale bar: 20 μ m. (C) Sequence differences in *rseA* gene among strains in clades C, F, and G. UTR regions are marked in pink, exons in purple, and nucleotide differences in white. (D) Colonies and hyphae stained with SYBR Green of RIB40 and the $\Delta rseA$ strain grown for 3 days on the minimal medium. Scale bar: 10 μ m. (E) Colonies and hyphae stained with SYBR Green of RIB915 expressing *rseA* gene from RIB40 and TK-47 strains with increased nuclei, and TK-41 strain without increased nuclei. Scale bar: 20 μ m. (F) Hyphal width of RIB915 and strains in E (n=14–23). Strains with *rseA* from strains with increased nuclei are marked in red, and those without are marked in blue. (G) Hyphal images stained with SYBR Green of *T. reesei* (QM9414) and *P. chrysogenum* (IFO4688) grown for 3 days on the minimal medium with or without 1% yeast extract. Scale bar: 20 μ m. (H) Ratio of class I-III hyphae in the *T. reesei* and *P. chrysogenum* control and bred strains cultured on the minimal medium with or without 1% yeast extract for 3 days (n=50). (I) CMCase activity per biomass of *T. reesei* industrial and control strains under the conditions in G (mean $\pm$ S.E., n=3, ** p < 0.01, t-test). (J) Protease activity per biomass of *P. chrysogenum* industrial and control strains under the conditions in G (mean $\pm$ S.E., n=3, ** p < 0.01, t-test).

and G indicates a few amino acid substitutions within the ORF, but no common mutations corresponded to the phenotype (Fig. 5C, Fig. S4B). Structural prediction suggests that the mutations are located inside the enzyme domain (Fig. S4C).

The disruption of the *rseA* in *A. oryzae* RIB40 resulted in significant growth delay, increased branching, and the absence of hyphae with increased nuclei (Fig. 5D, Fig. S4D). The *rseA* gene DNA fragments from the strains with increased nuclei (RIB40, TK-47) or from the strain without increased nuclei (TK-41) were amplified by PCR and introduced ectopically into RIB915 (Fig. 5E). The strains expressing *rseA* from RIB40 or TK-47 exhibited an increase in hyphal width and a corresponding increase in the proportion of hyphae with more nuclei, whereas the strain expressing *rseA* from TK41 showed no increase in hyphal width or nuclear number (Fig. 5E, Fig. S4E).

Industrial breeding strains from other genera

To investigate whether the phenomenon of increased nuclear numbers also applies to other industrial fungi, we compared the bred strain of *Trichoderma reesei* (QM9414) for cellulase production and its control strain (QM6a) (41), as well as the bred strain of *Penicillium chrysogenum* (IFO4688) for penicillin production and wild type strain (42). In minimal medium, no significant differences in hyphal width or nuclear number were observed between the bred and control strains, whereas with the addition of yeast extract, a notable increase in nuclear number and hyphal width was observed only in the bred strains (Fig. 5G, Fig. S5A-C). The addition of yeast extract significantly increased cellulase and protease activity only in the bred strain of *T. reesei* and *P. chrysogenum*, respectively (Fig. 5I, Fig. S5D). These results suggest that even in modern biotechnology, where high enzyme or compound producing strains are bred, strains with increased nuclear numbers were selected, like the traditional breeding practices of koji-fungi.

Discussion

In *A. oryzae* some strains, hyphae grow about three times thicker and contain roughly ten times more nuclei than the initial hyphae (Fig. 1). These thicker hyphae emerge from existing branches, where nuclear division is promoted, resulting in the formation of hyphae with an increased number of nuclei (Fig. 2). A clear correlation was observed between hyphal cell volume and nuclear number (Fig. 1G). This consistency aligns with findings in other organisms, from unicellular yeast to multicellular plants and animals, where the nuclear-to-cytoplasmic ratio is tightly regulated, allowing cell cycle progression only upon reaching a critical cell size (1–3). Both the G1/S and G2/M transitions are controlled by size checkpoints, with specific mechanisms differing between species (44, 45).

In *A. nidulans*, it is known that the duration of the G1 and G2 phases is regulated by temperature (46). While the mechanisms governing the G1/S phase transition remain unclear (47), the G2/M phase transition has been shown to involve the following: During the interphase of *A. nidulans*, NimX (CDK1) binds to NimE (cyclin B), and its activity is regulated through phosphorylation by AnkA (Wee1 kinase) and NimT (Cdc25 phosphatase) (48–50). The activated NimX-NimE complex subsequently dephosphorylates by NimT and activates NimA (Never-in-Mitosis A) protein kinase (51, 52). NimA is essential for the transition from G2 to mitosis, and its expression is tightly regulated during the cell cycle (52). The mRNA level of *nimA* and *nimT* begin to increase in G2, reach a plateau during late G2 and M phases, and are sharply degraded at the end of mitosis upon re-entry into interphase (52, 53), meaning high expression levels of *nimA* and *nimT* result in a longer G2 phase. In thick hyphae, *nimA* and *nimT* mRNA was hardly detected, suggesting that thick hyphae sense cell volume in some way and

subsequently shorten the G2 phase (**Fig. 6A**). The expression level of *nimE* was low in both thick and thin hyphae, with no significant difference observed. As known in other organisms, its function is likely regulated through phosphorylation and the protein degradation.

Branching is thought to occur through the degradation and reconstruction of the cell wall at the branching site (54). SNP analysis between industrial strains with different phenotypes identified RseA, a predicted glycosyltransferase (**Fig. 5**). Deletion of *rseA* caused defects in cell wall synthesis in *A. nidulans*, leading to increased enzyme secretion (40, 55). Although the targets of RseA are unknown, RseA could be involved in the cell wall integrity by modifying glycosyl chains on enzymes responsible for cell wall synthesis and degradation. Mutations in *rseA* may impair the regulation of cell wall loosening at the branching site, resulting in the formation of thicker hyphae. In fact, in the thick hyphae of RIB40, localized imbalances in cell wall components have been demonstrated by TEM and AFM.

Thicker hyphae can only form if high turgor pressure is maintained within the hyphae, in addition to changes in cell wall remodeling during branching. *msyA* and *msyB*, whose orthologues are calcium channels involved in the regulation of cell volume, were identified among the genes that show a significant increase in expression in thicker hyphae (**Fig. 4**) (38, 39). Analysis of the knockout strains demonstrated the function of cell volume regulation adapted to increased turgor pressure in hypoosmotic conditions. The function of these calcium channels is likely crucial for maintaining the balance of turgor pressure and cell volume.

The Target of Rapamycin (TOR) pathway is a central regulator of cellular growth, responding to a variety of nutrients, such as amino acids, and carbon sources, as well as environmental factors like oxygen levels, osmotic pressure, pH, and temperature stress (56, 57). Upon sensing these inputs, the TOR pathway influences translation activity, metabolic flow, and cell size (57, 58). In both yeast and mammalian models, amino acids like branched-chain amino acids (BCAAs), glutamine, and alanine are crucial for nutrient sensing, forming the basis for TOR signaling activation (58–60). These amino acids are like those contributing to nuclear increase in the RIB915 strain (**Fig. 3G**). Furthermore, TOR inhibition by rapamycin in the RIB40 strain suppressed the formation of hyphae with increased nuclei (**Fig. 3I–K**). Consistent with the activation of translation by the TOR pathway, the expression of ribosome-related genes was elevated in thicker hyphae (Fig. S3A). The TOR pathway directly regulates mitochondrial activity (61). Thick hyphae exhibited heightened mitochondrial density and activity (Fig. S1C, D), suggesting that the TOR pathway support the high ATP demand associated with rapid biosynthesis.

Cell volume and nuclear number mutually regulate each other, meaning that thicker hyphae always possess more nuclei. For the formation of thicker hyphae with numerous nuclei, an increase in cell volume and a corresponding increase in nuclei must occur simultaneously (**Fig. 6A**). In our model, cell wall loosening at a branching site and regulation of cell volume by turgor pressure constitute necessary conditions for increasing cell volume and maintaining thick hyphae. RseA and MsyA may be involved in these processes. At the same time, enhanced translational capacity, possibly due to increased expression of ribosomal genes associated with TOR activation, and mechanisms that accelerate the cell cycle represent another essential condition that enables an increase in nuclear number. Both genetic potential and nutritional environmental signals are likely required for the formation of thick hyphae with a high number of nuclei. In the *A. oryzae* RIB915 strain, the number of nuclei and cell volume increased in response to specific amino acids, whereas in the *A. oryzae* RIB40 and other strains, both increased even without the addition of such amino acids, suggesting that they possess the genetic potential to respond independently of nutritional signals. When thick hyphae were cultured on fresh medium, thin hyphae initially emerged, suggesting the necessity of sustained high metabolic activity. Weakening the cell wall by treatment with a low concentration of calcofluor white did not lead to hyphal thickening or an increase in nuclear number. On the contrary, thick hyphae have thicker cell walls (**Fig. 2H–K**), which are necessary to maintain the increased cell volume.

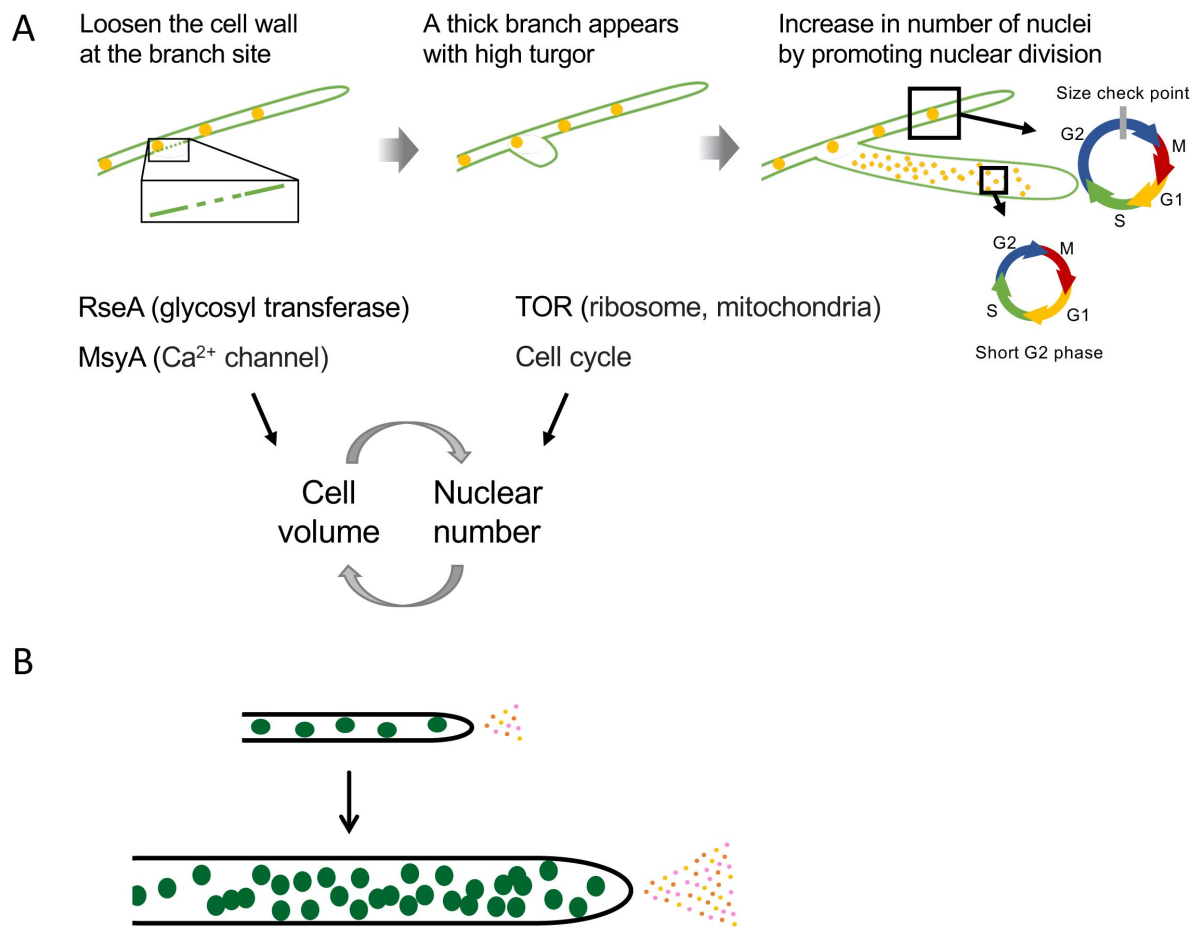


Figure 6.

(A) Molecular mechanism by which cell volume and nuclear number regulate each other and each increases simultaneously.
 (B) An increase in the number of nuclei per hyphal cell enhances transcription, translation, and enzyme secretion per cell.

A. nidulans and *A. flavus* did not exhibit an increase in nuclear number. *A. sojae* displayed increased nuclear numbers, whereas *A. kawachii* and *A. luchuensis* did not (Fig. 1). The 20 strains of *A. oryzae* selected from different clades showed an increase in nuclear number in the minimal medium supplemented with yeast extract. In the medium without yeast extract, there were strains that showed an increase in nuclear number and others that did not. These differences might be due to variations in their applications or the selection of strains with diverse characteristics, such as not only enzyme activity but also their impact on the flavor, aroma, and color of the final products. Additionally, there might be dominant mutations that activate TOR pathway even in the minimal medium. Multiple gene mutations were identified as contributing to the phenotype of increased hyphal volume and nuclear number, suggesting that the phenotype was not the result of a single mutation but rather the gradual accumulation of mutations through breeding. The koji-fungus strains used to produce high-quality sake and soy sauce have been selectively bred over many years, long before the advent of modern biotechnology, to enhance their saccharolytic and proteolytic activities. Modern biotechnological bred strains such as *T. reesei* QM9414 and *P. chrysogenum* IFO4688 also exhibit a phenotype of thicker hyphae with increased nuclear numbers under nutrient-rich conditions (Fig. 5). This consistency suggests that by breeding naturally multinucleate filamentous fungi under nutrient-rich conditions, it is possible to obtain strains with increased nuclear numbers and enhanced enzyme or compound production.

A previous study reported an increase in the number of nuclei in *A. nidulans* (62, 63). Here, we examined the nuclear distribution of *A. nidulans* grown on the culture media, however, did not find class III hyphae as observed in *A. oryzae*. Even in *A. nidulans*, conidiophore stalks contain a high number of nuclei. It has been shown that *A. oryzae* has a taller conidiophore stalk (64). In the thick hyphae of *A. oryzae*, the expression level of *flbA*, an early regulator of conidiophore development (65), was elevated. This suggests that differentiation to aerial hyphae may be involved in the increase of hyphal volume and nuclear number.

We investigated whether the reduction in nuclear size observed in thick hyphae was due to a change from diploid to haploid status. However, no difference in GFP-histone fluorescence intensity was detected between thick and thin hyphae (Fig. S5D). In both RIB40 and RIB915 strains, no significant difference in conidial spore size was observed despite the large difference in the number of nuclei within the hyphae (Fig. S5E). These results suggest that both thick and thin hyphae remain haploid, and that the smaller nuclear size observed in thick hyphae is likely due to a higher nuclear density.

The number of nuclei in filamentous fungi varies greatly among species (66). For example, *Neurospora crassa* can exceed 100 nuclei, which correlates with its exceptionally rapid growth rate and high protein secretion capacity (67). An increase in the number of nuclei per cell is expected to enhance transcription and translation per cell, thereby improving enzyme production and secretion capacity (Fig. 6B). Whereas, maintaining a high nuclear number requires a substantial amount of energy, and in conditions where external nutrient supply is insufficient, cells rapidly die. Thus, the tendency to maintain a high nuclear number could be a selective pressure unlikely to occur in nature. The regulation of nuclear number and its ecological strategy are intriguing in other fungi such as *N. crassa*, which rapidly spreads after wildfires (68), and arbuscular mycorrhiza fungi that form symbiotic relationships with plants and contain thousands of nuclei within hyphae lacking septa (69). In any case, koji-fungus strains were bred specifically for high enzyme production in nutrient-rich environments, and we propose that some of them acquired the specialized trait of increased cell volume and nuclear number through selection. This study elucidates the evolutionary processes driven by breeding selection in *A. oryzae* and the underlying mechanisms contributing to high enzyme productivity. These findings not only offer new approaches for optimizing filamentous fungi for bioindustry applications but also lead to the elucidation of the molecular mechanisms regulating cell volume and nuclear number in multinucleated organisms.

Methods

Strains and Culture Conditions

The filamentous fungal strains used in this study are listed in [Table S6](#). *A. oryzae* RIB40 expressing H2B-GFP, UtH2BG strain, is described previously ([33](#)). Minimal medium for *Aspergillus* species was shown in [Table S7](#). When peptone, casamino acids, or yeast extract was added, the final concentration was 1%. Amino acids were added at a final concentration of 0.1%, B vitamins (biotin, pyridoxine hydrochloride, thiamin hydrochloride, riboflavin, PABA, and nicotinic acid) were added at 100 μ M, and nucleotides (inosine and uridine) were added at 10 mM.

Genetic Manipulation

To delete target genes, genome editing plasmids and circular donor DNA plasmids were constructed as described previously ([70](#)). The sgRNA cassette was designed under the U6 promoter with the following target sequences: GTGACGATCTACGGTAACGC for the *msyA* gene, GATCAGTTGAGCGTCCCCTA for the *msyB* gene, and AGCTAGCGCAGCCGGTCTGC for the *rseA* gene. The genome editing plasmid was constructed by introducing the sgRNA expression cassette into the SmaI restriction site of the pRGE-gRT6 plasmid ([70](#)). For the donor plasmid, 1 kb fragments upstream and downstream of the target genes were amplified, ligated, and inserted into the linearized pUC19 vector (Takara Bio). Both plasmids were applied for transformation using the *A. oryzae* RIB40 strain. To amplify *rseA* region, spanning 1 kb upstream to 1 kb downstream, the templates used were *A. oryzae* RIB40, TK-42, and TK-47 strains. The amplified fragments were inserted into the HindIII site of the pPTRI vector (TaKaRa) to construct the plasmid. Transformations were performed using the protoplast-PEG method according to the manufacturer's protocol for pPTRI. Transformants were selected on the minimal medium plates containing 0.1 mg/l pyrithiamine. Integration of constructs was confirmed by PCR analysis.

Microscopy

Images were captured using an Axio Observer Z1 inverted epifluorescence microscope (Carl Zeiss) equipped with Plan-Apochromat objectives, an AxioCam 506 mono camera, and a Colibri.2 LED light source. The stage temperature was maintained at 30°C with a thermoplate (Tokai Hit). Nuclei were stained with SYBR Green (Takara), mitochondria with Rhodamine 123 (Invitrogen), the cell membrane with FM4-64 (Invitrogen). Images were collected using Zen (Carl Zeiss) and ImageJ software.

Image Analysis

The distribution pattern of nuclei was classified into the following three types; Class I: nuclei distribute at a constant interval without overlapping; Class II: nuclei align but sometimes overlap; Class III: nuclei scattered throughout hyphae but not aligned. Nuclei, hyphal size, and fluorescence intensity were quantified using Zen and ImageJ software. Z-stack 3D reconstruction and nuclear color coding were performed with Imaris (Oxford Instruments). Hyphal elongation distances were analyzed using the MTrackJ plugin in ImageJ. For [Fig. 2E](#), nuclei were binarized, and those in non-increased hyphae were displayed in white based on size and circularity.

Transmission Electron Microscopy (TEM)

The culture samples were embedded in 1% water agar and fixed with 2.5% glutaraldehyde in 0.1 M phosphate buffer (pH 7.0) at room temperature for 2 hours. Sample preparation and sectioning were performed at the Institute of Medicine, University of Tsukuba. Observations were conducted using a JEM-1400 transmission electron microscope (JEOL).

Imaging with Microfluidic Devices

Microfluidic devices were fabricated as described previously (33). Two types of devices were constructed: a 2D observation device to visualize hyphal expansion (Fig. S1A) and a single-hypha isolation device for enzymatic activity measurement (Fig. S2A). Conidia suspensions were loaded into a 10 ml plastic syringe (SS-10ESZ, Terumo) and connected to polyethylene tubing (inner diameter 0.38 mm, outer diameter 1.09 mm, BD Intramedic). Air was expelled from the syringe, and the suspension was injected into the device inlet. The number of conidia flowing through the microfluidic channel was adjusted under a microscope. The syringe was then replaced with another containing medium, and the medium was delivered at the minimum flow rate using a syringe pump (YSP-101, YMC). The device was incubated at 30°C on a thermal stage (TOKAI HIT) and imaged using an Axio Observer Z1 microscope (Carl Zeiss). α -amylase activity in the device was measured using the EnzChek Ultra amylase assay kit (Invitrogen).

Enzyme Activity Assays

Fungal conidia 10^4 were inoculated onto 25 ml of medium in a petri dish and statically cultured at 30 °C for 4 days. Protein concentration and enzymatic activities in the culture supernatant were measured. Protein quantification was performed using the Bradford assay (Protein Assay Kit, BIO-RAD), with bovine gamma globulin as the standard. Carboxymethyl cellulase (CMCase) activity was determined by measuring reducing sugars using the 3,5-dinitrosalicylic acid method. CMCase activity was assayed at 50 °C for 15 min in a reaction mixture containing 50 mM sodium acetate buffer (pH 5.0) and 1% carboxymethyl cellulose. One unit of enzyme activity was defined as the amount of enzyme required to release 1 μ mol of reducing sugar (as glucose equivalent) per minute. α -amylase activity was measured using an α -Amylase Assay Kit (Kikkoman Biochemifa) according to the manufacturer's instructions. Acid carboxypeptidase (ACP) activity was determined using the Acid Carboxypeptidase Assay Kit (Kikkoman Biochemifa) following the provided protocol.

RNA-seq Analysis

Sterilized cellophane was placed on agar plates, and conidia were inoculated. After culturing at 30 °C for 3 days, samples were frozen, homogenized with a mortar and pestle, and total RNA was extracted using the RNeasy Mini Kit (QIAGEN). Library preparation, sequencing, and partial data analysis were performed by the Faculty of Medicine, University of Tsukuba. Sequencing was conducted using 36 bp paired end reads on the Illumina NextSeq 500, and reads were mapped to the *A. oryzae* RIB40 reference genome using the CLC Genomic Workbench (QIAGEN). For laser microdissection, sterilized membrane slides (Carl Zeiss) were immersed in MM liquid, and conidia were inoculated onto the membrane. After static culturing at 30 °C for 3 days, sections were collected using a PALM MicroBeam laser microdissection system (Carl Zeiss). RNA was extracted from 500-700 sections respectively using the RNeasy Mini Kit. Library preparation and sequencing were performed by Takara Bio Inc., using 150 bp paired end reads on the Illumina NovaSeq 6000. Reads were mapped to the *A. oryzae* RIB40 reference genome using the CLC Genomic Workbench. Genes showing a ≥ 4 -fold change in expression were considered significant, and gene ontology (GO) enrichment analysis of biological processes was performed using ShinyGO 0.81 (71). Heatmaps were visualized using the Python Seaborn package (72). Gene annotations were obtained from FungiDB.

Low Osmotic Pressure Shock Analysis

Conidia were spot inoculated onto the minimal medium containing 1 M NaCl and cultured at 30 °C for 3 days. Colony tips were treated with distilled water for 3 min, and the number of tip ruptures was counted. Hyphal tip swelling was assessed by staining hyphae cultured in the minimal medium liquid for 3 days with FM4-64, mounting them on a coverslip with a small volume of medium, and adding an equal volume of distilled water.

Raman Spectroscopy

The conidia were inoculated into minimal medium liquid in a glass-bottom dish and cultured at 30 °C for 3 days. Raman spectra were acquired using an XploRA PLUS Raman microscope, as described previously (32 [↗](#)). Hyphae attached to the dish bottom were irradiated with a 532 nm laser at 0.93 mW for 30 seconds. Spike artifacts caused by cosmic rays were manually removed. Ten spectra were averaged, and background signals attributed to the medium and glass were subtracted.

Protein Structure Analysis

RseA protein structure was predicted using AlphaFold2 and visualized with PyMOL (v1.20). Domains were annotated using InterPro.

Atomic Force Microscopy (AFM)

The conidia were inoculated into minimal medium liquid on a Poly-L-Lysine-coated (Sigma Aldrich) glass-bottom dish (ibidi) and cultured at 30 °C for 3 days. We used a JPK Nanowizard 4 (Bruker) equipped with an inverted fluorescence microscope (Eclipse Ti2, Nikon). The temperature was maintained at 30 °C using a dish heater integrated into the stage. BL-AC40TS-C2 cantilevers (Olympus, spring constant approximately 0.1 N/m) were used for AFM imaging under the following conditions: QI mode, $1.5 \times 1.5 \mu\text{m}$ or $0.5 \times 0.5 \mu\text{m}$ scan size, 128×128 pixels, Z-length $1 \mu\text{m}$, setpoint 0.1 nN, and Z speed $166 \mu\text{m/s}$. The Young's modulus was calculated using JPKSPM Data Processing software (Bruker). After averaging and baseline subtraction, Young's modulus was calculated using a Hertz model with a triangular pyramid half-cone angle of 35°.

SNP analysis

The genome data of *A. oryzae* required for SNP identification was obtained from the National Center for Biotechnology Information (NCBI), and mapped to *A. oryzae* RIB40 reference sequence using BWA-MEM2 (73 [↗](#)). The SNPs were listed by FreeBayes (74 [↗](#)), and detailed comparative analysis of genomes was performed on Integrative Genome Viewer (IGV) manually.

RNA seq

The RNA seq data have been deposited in DDBJ as BioProject PRJDB19992.

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Additional files

Supplementary materials, Fig. S1-5, Table S1-7 legends, Movie 1-6 legends [↗](#)

Table S1 [↗](#)

Table S2 [↗](#)

Table S3 [↗](#)

Table S4 [↗](#)

Table S5 [↗](#)

Table S6 [↗](#)

Table S7 [↗](#)

Movie 1 [↗](#)

Movie 2 [↗](#)

Movie 3 [↗](#)

Movie 4 [↗](#)

Movie 5 [↗](#)

Movie 6 [↗](#)

Additional information

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Reviewer #1 (Public review):

Filamentous fungi are established work horses in biotechnology with *Aspergillus oryzae* as a prominent example with a thousand-year of history. Still the cell biology and biochemical properties of the production strains is not well understood. The paper of the Takeshita group describes the change in nuclear numbers and correlate it to different production capacities. They used microfluidic devices to really correlate the production with nuclear numbers. In addition, they used microdissection to understand expression profile changes and found an increase of ribosomes. The analysis of two genes involved in cell volume control in *S. pombe* did not reveal conclusive answers to explain the phenomenon. It appears that it is a multi-trait phenotype. Finally, they identified SNPs in many industrial strains and tried to correlate them to the capability of increasing their nuclear numbers.

The methods used in the paper range from high quality cell biology, Raman spectroscopy to atomic force and electron microscopy and from laser microdissection to the use of microfluidic devices to study individual hyphae.

This is a very interesting, biotechnologically relevant paper with the application of excellent cell biology.

Comments on revised version:

The authors addressed all suggestions satisfactorily.

<https://doi.org/10.7554/eLife.107043.2.sa3>

Reviewer #2 (Public review):

Summary:

In the study presented by Itani and colleagues it is shown that some strains of *Aspergillus oryzae* - especially those used industrially for the production of sake and soy sauce - develop hyphae with a significantly increased number of nuclei and cell volume over time. These thick hyphae are formed by branching from normal hyphae and grow faster and therefore dominate the colonies. The number of nuclei positively correlates with the thicker hyphae

and also the amount of secreted enzymes. The addition of nutrients such as yeast extract or certain amino acids enhanced this effect. Genome and transcriptome analyses identified genes, including *rseA*, that are associated with the increased number of nuclei and enzyme production. The authors conclude from their data involvement of glycosyltransferases, calcium channels and the tor regulatory cascade in regulation of cell volume and number of nuclei. Thicker hyphae and an increased number of nuclei was also observed in high-production strains of other industrially used fungi such as *Trichoderma reesei* and *Penicillium chrysogenum*, leading to the hypothesis that the mentioned phenotypes are characteristic of production strains which is of significant interest for fungal biotechnology.

Strengths:

The study is very comprehensive and involves application of diverse state-of-the-art cell biological, biochemical and genetical methods. Overall, the data are properly controlled and analyzed, figures and movies are of excellent quality.

The results are particularly interesting with regard to the elucidation of molecular mechanisms that regulate the size of fungal hyphae and their number of nuclei. For this, the authors have discovered a very good model: (regular) strains with a low number of nuclei and strains with high number of nuclei. Also, the results can be expected to be of interest for the further optimization of industrially relevant filamentous fungi.

In the revision the authors addressed all my comments and as a result produced an even stronger study.

<https://doi.org/10.7554/eLife.107043.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors seek to determine the underlying traits that support the exceptional capacity of *Aspergillus oryzae* to secrete enzymes and heterologous proteins. To do so, they leverage the availability of multiple domesticated isolates of *A. oryzae* along with other *Aspergillus* species to perform comparative imaging and genomic analysis.

Strengths:

The strength of this study lies in the use of multifaceted approaches to identify significant differences in hyphal morphology that correlate with enzyme secretion, which is then followed by the use of genomics to identify candidate functions that underlie these differences.

Weaknesses:

Although the image analysis and data interpretation is convincing, the genetic data supporting the author's model is somewhat more speculative and will likely require additional investigation.

Overall, the authors have achieved their aims in that they are able to clearly document the presence of two distinct hyphal forms in *A. oryzae* and other *Aspergillus* species, and to correlate the presence of the thicker rapidly growing form with enhanced enzyme secretion. The image analysis is convincing. The discovery that addition of yeast extract and specific amino acids can stimulate formation of the novel hyphal form is also notable. Although the conclusions are generally supported by the results, this is perhaps less so for the genetic analysis as it remains unclear how direct the role of *RseA* and the calcium transporters might be in supporting the formation of the thicker hyphae.

The results presented here will impact the field. The complexity of hyphal morphology and how it affects secretion are not well understood despite the importance of these processes for the fungal lifestyle. In addition, the description of approaches that can be used to facilitate the study of these different hyphal forms (i.e., stimulation using yeast extract or specific amino acids) will benefit future efforts to understand the molecular basis of their formation.

<https://doi.org/10.7554/eLife.107043.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

*Filamentous fungi are established workhorses in biotechnology, with *Aspergillus oryzae* as a prominent example with a thousand-year history. Still, the cell biology and biochemical properties of the production strains is not well understood. The paper of the Takeshita group describes the change in nuclear numbers and correlates it to different production capacities. They used microfluidic devices to really correlate the production with nuclear numbers. In addition, they used microdissection to understand expression profile changes and found an increase in ribosomes. The analysis of two genes involved in cell volume control in *S. pombe* did not reveal conclusive answers to explain the phenomenon. It appears that it is a multi-trait phenotype. Finally, they identified SNPs in many industrial strains and tried to correlate them to the capability of increasing their nuclear numbers.*

The methods used in the paper range from high-quality cell biology, Raman spectroscopy, to atomic force and electron microscopy, and from laser microdissection to the use of microfluidic devices to

study individual hyphae.

This is a very interesting, biotechnologically relevant paper with the application of excellent cell biology. I have only minor suggestions for improvement.

We sincerely appreciate your fair and positive evaluation of our work. Thank you for your suggestions for improvement. We respond to each of them appropriately.

Reviewer #2 (Public review):

Summary:

*In the study presented by Itani and colleagues, it is shown that some strains of *Aspergillus oryzae* - especially those used industrially for the production of sake and soy sauce - develop hyphae with a significantly increased number of nuclei and cell volume over time. These thick hyphae are formed by branching from normal hyphae and grow faster and therefore dominate the colonies. The number of nuclei positively correlates with the thicker hyphae and also the amount of secreted enzymes. The addition of nutrients such as yeast extract or certain amino acids enhanced this effect. Genome and transcriptome analyses identified genes, including *rseA*, that are associated with the increased number of nuclei and enzyme production. The authors conclude from their data involvement of glycosyltransferases, calcium channels, and the tor regulatory cascade in the regulation of cell volume and number of nuclei. Thicker hyphae and an increased number of nuclei were also observed in high-production strains of other industrially used fungi such as *Trichoderma reesei* and *Penicillium chrysogenum*, leading*

to the hypothesis that the mentioned phenotypes are characteristic of production strains, which is of significant interest for fungal biotechnology.

Strengths:

The study is very comprehensive and involves the application of diverse state-of-the-art cell biological, biochemical, and genetic methods. Overall, the data are properly controlled and analyzed, figures and

movies are of excellent quality.

The results are particularly interesting with regard to the elucidation of molecular mechanisms that regulate the size of fungal hyphae and their number of nuclei. For this, the authors have discovered a very good model: (regular) strains with a low number of nuclei and strains with a high number of nuclei. Also, the results can be expected to be of interest for the further optimization of industrially relevant filamentous

fungi.

Weaknesses:

There are only a few open questions concerning the activity of the many nuclei in production strains (active versus inactive), their number of chromosomes (haploid/diploid), and whether hyper-branching always leads to propagation of nuclei.

We are very grateful for your recognition of our findings, the proposed model, and their significance for future applications. We are grateful for the questions, which contribute to a more accurate understanding.

Our responses to each are provided below.

Reviewer #3 (Public review):

Summary:

*The authors seek to determine the underlying traits that support the exceptional capacity of *Aspergillus oryzae* to secrete enzymes and heterologous proteins. To do so, they leverage the availability of multiple domesticated isolates of *A. oryzae* along with other *Aspergillus* species to perform comparative imaging and genomic analysis.*

Strengths:

The strength of this study lies in the use of multifaceted approaches to identify significant differences in hyphal morphology that correlate with enzyme secretion, which is then followed by the use of genomics to identify candidate functions that underlie these differences.

Weaknesses:

There are aspects of the methods that would benefit from the inclusion of more detail on how experiments were performed and data interpreted.

*Overall, the authors have achieved their aims in that they are able to clearly document the presence of two distinct hyphal forms in *A. oryzae* and other *Aspergillus* species, and to correlate the presence of the thicker, rapidly growing form with enhanced enzyme secretion. The image analysis is convincing. The discovery that the addition of yeast extract and specific amino acids can stimulate the formation of the novel hyphal form is also notable. Although the conclusions are generally supported by the results, this is*

perhaps less so for the genetic analysis as it remains unclear how direct the role of RseA and the calcium transporters might be in supporting the formation of the thicker hyphae.

The results presented here will impact the field. The complexity of hyphal morphology and how it affects secretion is not well understood despite the importance of these processes for the fungal lifestyle. In addition, the description of approaches that can be used to facilitate the study of these different hyphal forms (i.e., stimulation using yeast extract or specific amino acids) will benefit future efforts to understand the molecular basis of their formation.

We are very grateful for your fair and thoughtful evaluation of our work. We agree that the genetic analysis in the latter part is relatively weaker compared to the imaging analysis in the first half. Rather than a single mutation causing a dramatic phenotypic change, we believe that the accumulation of various mutations through breeding leads to the observed phenotype, making it difficult to clearly demonstrate causality. Since transcriptome and SNP analyses have revealed key pathways and phenotypes, it would be gratifying if these insights could contribute to future applications utilizing filamentous fungi.

Reviewer #1 (Recommendations for the authors):

I was wondering what happens if thick hyphae were taken as inoculum for a new colony or thin hyphae. Is it possible to enrich for one or the other type of hyphae? Perhaps in the presence of yeast extract or certain amino acids.

Added an explanation in the discussion.

L304-306. When thick hyphae were cultured on fresh medium, thin hyphae initially emerged, suggesting that sustained metabolic activity is required for the formation of thick hyphae with a high number of nuclei.

L120-121. In some cases, thick hyphae emerged by branching from thick hyphae (Fig. 2D, left), while in other cases, thin hyphae emerged from thick hyphae (Fig. 2D, right). Thin hyphae emerge in the early stage of cultivation even in the presence of yeast extract or certain amino acids.

*In the Discussion, they hypothesize that the primary effect could be on cell wall rigidity. I am wondering if that hypothesis could be tested by adding, for instance, sublethal concentrations of cytochalasin to hyphae of *A. nidulans* to weaken the cell wall.*

The question is reasonable. To ensure accurate understanding, we moved Fig. S6 to Fig. 6 and revised the discussion as follows.

L294-295. In our model, cell wall loosening at a branching site and regulation of cell volume by turgor pressure constitute necessary conditions for increasing cell volume and maintaining thick hyphae. L306-309. Weakening the cell wall by treatment with a low concentration of calcofluor white did not lead to hyphal thickening or an increase in nuclear number. On the contrary, thick hyphae have thicker cell walls (Fig. 2H-K), which are necessary to maintain the increased cell volume.

*I recommend including some older literature. It was described already 20 years ago that *A. niger* differentiates hyphae with different capacities to secrete proteins (PMID: 16238620). In addition, there are old reports in *A. nidulans* reporting high numbers of nuclei (<https://doi.org/10.1099/00221287-60-1-133>). Perhaps it is worth trying to reproduce those cultural conditions. At least this should be discussed. In the same line, the number of nuclei increases a lot in the stalk of conidiophores in *A. nidulans*. These*

observations could be used as examples that the phenomenon observed in A. oryzae may be of general importance.

Thank you for the suggestion. It is a very interesting proposal. We checked the nuclei distribution of *A. nidulans* on the media and added the following discussion.

L328-334. A previous study reported an increase in the number of nuclei in *A. nidulans* (62, 63). Here, we examined the nuclear distribution of *A. nidulans* grown on the culture media, however, did not find class III hyphae as observed in *A. oryzae*. Even in *A. nidulans*, conidiophore stalks contain a high number of nuclei. It has been shown that *A. oryzae* has a taller conidiophore stalk (64). In the thick hyphae of *A. oryzae*, the expression level of *flbA*, an early regulator of conidiophore development (65), was elevated. This suggests that differentiation to aerial hyphae may be involved in the increase of hyphal volume and nuclear number.

(62) Clutterbuck A.J. Synchronous Nuclear Division and Septation in *Aspergillus nidulans*. J Gen Microbiol 60, 133-135 (1970).

(63) Vinck, A., Terlouw, M., Pestman, W.R., Martens, E.P., Ram, A.F., van den Hondel, C.A., Wösten, H.A. Hyphal differentiation in the exploring mycelium of *Aspergillus niger*. Mol Microbiol 58, 693-9 (2005).

(64) Wada R, Maruyama J, Yamaguchi H, Yamamoto N, Wagu Y, Paoletti M, Archer DB, Dyer PS, Kitamoto K. Presence and functionality of mating type genes in the supposedly asexual filamentous fungus *Aspergillus oryzae*. Appl Environ Microbiol 78, 2819-29 (2012).

(65) Lee, B.N., Adams, T.H. Overexpression of *flbA*, an early regulator of *Aspergillus* asexual sporulation, leads to activation of *brlA* and premature initiation of development. Mol Microbiol 14, 323-34 (1994).

Reviewer #2 (Recommendations for the authors):

I suggest addressing the following questions to strengthen the manuscript:

(1) Do the authors have an explanation for their result that with an increase in the number of nuclei the individual nucleus is smaller? Have the authors checked whether all the nuclei are haploid or diploid?

Thank you for the very important question. We added new results to Fig. S5D and S5E and the following discussion.

L335-340. We investigated whether the reduction in nuclear size observed in thick hyphae was due to a change from diploid to haploid status. However, no difference in GFP-histone fluorescence intensity was detected between thick and thin hyphae (Fig. S5D). In both RIB40 and RIB915 strains, no significant difference in conidial spore size was observed despite the large difference in the number of nuclei within the hyphae (Fig. S5E). These results suggest that both thick and thin hyphae remain haploid, and that the smaller nuclear size observed in thick hyphae is likely due to a higher nuclear density.

(2) In this context, the biological relevance of the increase in the number of nuclei should also be discussed in more detail. It remains to be clarified whether in hyphae with a high number of nuclei all nuclei are functionally active or whether many nuclei are possibly "inactive". Studies on the transcriptional activity of individual nuclei or on DNA replication (e.g., by EdU labeling) could clarify this.

Added the explanation below.

L102-105. The transcriptional activity of each nucleus is unknown. However, a previous study (Yasui et al., FBB 2020) demonstrated that nuclear division is synchronized even when there are more than 200 nuclei. This suggests that DNA replication occurs similarly in most nuclei. Furthermore, since the germination rate of conidia and the colonies formed from individual conidia show no significant abnormalities, it is suggested that nearly all nuclei possess normal genomes and chromosomes.

(3) It becomes not entirely clear what the underlying signal is that causes a thin hypha to branch into a thick multinucleated cell. This needs to be discussed in more detail.

Thanks for the suggestion. We clarified the signal to increase nuclear number and cell volume.

L294-309. Although it is speculative, we propose a model to aid interpretation in the discussion. We have clarified that both genetic potential and environmental signals such as nutrients are important.

(4) Is increased branching always correlated with an increased number of nuclei?

It is not an increase in branching, but rather the thickening of hyphae and an increase in cell volume that is consistently associated with an increase in nuclear number. Approximately 40 hours after inoculation, within 400 μm from the tip, the number of branches was 3.4 (SD=2.4) in thin hyphae and 2.6 (SD=0.5) in thick hyphae, suggesting that branching does not increase (n=4). Since thick hyphae elongate faster, it seems that fewer branches are present near the tip, even if the branching frequency itself remains unchanged.

(5) The abstract does not summarize the many findings of the manuscript in an adequate way.

abstract change

Minor:

(1) Lines 49-50: Why italics?

corrected.

(2) Line 179: process.

corrected.

(3) Lines 313-314: Do not forget (and discuss) in this context mycorrhiza fungi with up to thousands of nuclei that were apparently selected during evolution for this high number of nuclei.

Thank you for the very interesting suggestion. We have added the following discussion.

L339-351. The regulation of nuclear number and its ecological strategy are intriguing in other fungi such as *N. crassa*, which rapidly spreads after wildfires (68), and arbuscular mycorrhiza fungi that form symbiotic relationships with plants and contain thousands of nuclei within hyphae lacking septa (69).

(68) Jacobson, D. J. et al. *Neurospora* in temperate forests of western North America. *Mycologia* 96, 66–74 (2004).

(69) Kokkoris V, Stefani F, Dalpé Y, Dettman J, Corradi N. Nuclear Dynamics in the Arbuscular Mycorrhizal Fungi. *Trends Plant Sci.* 25, 765-778 (2020).

| (4) Lines 356-358: *many typos.*

corrected.

| **Reviewer #3 (Recommendations for the authors):**

Specific suggestions or clarifications for the authors include:

| (1) Lines 49-50: *Is this sentence italicized for a reason?*

It was a mistake, so we have corrected it.

| (2) Line 83: *More detail on the specific characteristics of the different classes of hyphae would be helpful. Perhaps include a schematic drawing that emphasizes the differences between class I,II, and III hyphae.*

L398-400. The classification is described in the Methods section: Class I – nuclei are distributed at regular intervals without overlapping; Class II – nuclei are aligned but occasionally overlap; Class III – nuclei are scattered throughout the hyphae without alignment. Representative images are shown in a previous study (Yasui et al., FBB 2020).

L82-84. We have added this information to clarify the classification.

| (3) Lines 102-103: *It was not very clear how this experiment was done. Are you counting nuclei within 100 μ m of the tip? Are these all in one hyphal compartment? These details could be provided in a drawing that would make it easier for the reader to understand how this was done.*

L109. Due to variation in the distance from the hyphal tip to the septum, we counted the number of nuclei within 100 μ m from the hyphal tip. When septa were present, nuclei were counted in the same manner, so multiple compartments may be included. Changed the explanation.

| (4) Lines 134-140: *Is there a way to calibrate levels of secreted protein or amylase activity per nucleus? That is, if the ratio of cytoplasmic volume per nucleus is constant, does the same apply to the secreted product? Knowing this would help to clarify whether the key feature in enhanced secretion is nuclear (e.g., gene expression) versus a cytoplasmic trait (e.g., vesicle trafficking).*

Enzyme activity was measured across the entire mycelium, which includes a mixture of hyphae with high and low numbers of nuclei. Therefore, it is difficult to assess the correlation between enzyme activity and nuclear number. Enzyme activity was normalized by fungal biomass. The size of each colony is shown in Fig. 1B. Additionally, the correlation between the proportion of hyphae with increased nuclear number and enzyme activity is shown in Fig. 3H. In the experiment where enzyme activity was measured in a single hypha, we attempted to measure the number of nuclei; however, we could not use the nuclear GFP strain because the substrate exhibits green fluorescence. DAPI staining also failed due to limited dye access to the microfluidic channel. Changed the section title, 'Increase in nuclear number and enzyme secretion' from 'Correlation between nuclear number and enzyme secretion'.

(5) Line 151 and Figure 3F: YE also triggered a ~5-fold enhancement of secretion in *A. nidulans* without a concomitant increase in hyphal width. This merits some comment in the text.

Added an explanation, L156-157.

In *A. nidulans*, the addition of yeast extract did not cause a dramatic increase in nuclear number, but hyphal width increased by 1.4-times and protein secretion increased by 5.1-times.

(6) Line 252: Were *nimE* levels detected or altered in thick hyphae? The levels of this cycling might play a more important role in a shortened cell cycle than the authors have considered, especially as *NimE* functions during both G1 and G2.

Added an explanation below, L260-262.

The expression level of *nimE* (AO090003000993) was low in both thick and thin hyphae, with no significant difference observed. As known in other organisms, its function is likely regulated through phosphorylation and the protein degradation.

(7) Line 254: Please provide a citation for the statement that branches emerge as a result of cell wall loosening.

rephrased and added citation, L263.

Branching is thought to occur through the degradation and reconstruction of the cell wall at the branching site (54).

Harris SD. Branching of fungal hyphae: regulation, mechanisms and comparison with other branching systems. *Mycologia* 100, 823-32 (2008).

(8) Lines 275-277: It would be interesting to know whether the addition of rapamycin also suppressed the ability of amino acids to trigger greater numbers of class III hyphae.

We added new results at Fig. S2G.

L168. Rapamycin decreased the ratio of hyphae with increased nuclei even in the medium with yeast extract (Fig. S2G).

(9) Lines 282-289: My sense is that this model is too speculative at this time. The role of *RseA* seems very broad based on the strong deletion phenotype. How would the removal of *RseA* be regulated to limit its effect to the branch site? Also, the *msyA* deletion phenotype isn't entirely consistent with what you would expect if it were necessary to maintain thick hyphae. Lastly, the authors do not show that translational capacity is enhanced in thick hyphae. I would suggest that these statements be tempered to some degree.

Thank you for your comment. We agree that it was too speculative, whereas we believe that some explanatory interpretation is necessary. Therefore, we have revised the text as follows, L294-300. In our model, cell wall loosening during branching and regulation of cell volume by turgor pressure constitute necessary conditions for increasing cell volume and maintaining thick hyphae. *RseA* and *MsyA* may be involved in these processes. At the same time, enhanced translational capacity by increased expression of ribosomal genes, possibly due to associated with TOR activation by specific amino acids, and mechanisms that accelerate the cell cycle represent another essential condition that enables an increase in nuclear number.

(10) General: how do the authors reconcile the observation that YE and amino acids stimulate the formation of thicker hyphae, yet the time lapse imaging (Figure 2E) suggests that these hyphae arise at a later time during colony development when these resources might be limiting? The authors should consider providing some insight into this in the Discussion.

L300-305. Added a discussion below.

Both genetic potential and nutritional environmental signals are likely required for the formation of thick hyphae with a high number of nuclei. When thick hyphae were cultured on fresh medium, thin hyphae initially emerged, suggesting the necessity of sustained high metabolic activity.

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